

Cybersecurity Incident Report:

SYN Flood

Section 1: Identify and Explain the Network Disruption

Our investigation into the network interruption began with an automated alert, signaling a problem with the web server and leading to a connection timeout error for website visitors.

Analyzing the TCP/HTTP logs, our cybersecurity analyst detected a surge in TCP SYN requests coming from an unfamiliar IP address, which is indicative of a potential Denial of Service (DoS) SYN flood attack.

Section 2: Explain How the Attack Is Causing the Website to Malfunction

In a SYN flood attack, the malicious actor exploits the TCP handshake by flooding the web server with connection requests it never intends to complete. Under normal conditions, the handshake follows three steps:

1. **SYN (Synchronize):** The initiating device sends a SYN packet to the destination device, indicating it wants to establish a connection.
2. **SYN-ACK (Synchronize-Acknowledge):** The destination device replies with a SYN-ACK packet, acknowledging the request and confirming it is ready to proceed.
3. **ACK (Acknowledge):** The initiating device sends a final ACK packet, completing the connection so data exchange can begin.

With a limited number of ports available, the server becomes overwhelmed trying to respond to the flood of half-open connections, leading to a slowdown and eventually a full operational failure. The impact includes lost revenue, damaged customer trust, and potential harm to server infrastructure or data.

To reduce the risk of this happening again, the following measures are recommended:

- Deploy a Next Generation Firewall (NGFW) to monitor network traffic for early warning signs.
- Use VPNs and encryption to help shield the web server's IP address.
- Segment the network into subnets so that an outage is contained rather than spreading across the whole infrastructure.